



Context-dependent leadership in global health science: Brazil's role across North–South, South–South, and triangular collaboration models (2010–2019)

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Received: 2 March 2026 / Accepted: 10 August 2026
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Abstract

Brazil plays an important role in global scientific collaboration in health and life sciences (HLS), yet systematic analysis of how patterns differ across North–South, South–South, and Triangular partnerships remains limited. This study examines Brazil's international HLS collaborations (2010–2019) using Scopus data (362,860 publications), analyzing growth, leadership, citation impact, and thematic specialization across three models: Brazil with high-income countries (BR–HIC; North–South), Brazil with low- and middle-income countries (BR–LMIC; South–South), and Brazil with both (BR–HIC–LMIC; triangular). International collaborations nearly doubled (1.71-fold), with Triangular cooperation growing fastest (3.08-fold), though this growth was not accompanied by a proportional increase in Brazilian authorship leadership. Scientific leadership, operationalized here as first and last authorship position, varied markedly across models: 75% in BR–LMIC, 67% in BR–HIC, and only 31% in Triangular partnerships, a gradient suggesting context-dependent rather than uniformly peripheral or central positioning. Citation impact followed a clear hierarchy: 49% of Triangular collaborations ranked in top 10% most-cited versus 34% for BR–HIC and 22% for BR–LMIC. Thematic focus diverged across models: BR–HIC spanned several areas and Triangular partnerships concentrated on clinical trials. Notably, BR–LMIC thematic focus aligned closely with Brazil's highest Specialization Index fields: Agricultural and Biological Sciences and Veterinary Sciences. These findings demonstrate that emerging scientific powers may occupy context-dependent positions in global research networks, with implications for how bibliometric frameworks assess leadership and equity in international collaboration.

Keywords Brazil · Scientific collaboration · Health research · South–South · North–South · Triangular

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Introduction

International collaboration has become a central focus in science and technology studies, reflecting its increasing importance for scientific productivity, citation impact, funding and infrastructure access, and knowledge sharing (Royal Society, 2011; Yao, 2021). In the health and life sciences (HLS), international collaboration plays a strategic role, as research outcomes often address transnational challenges, such as emerging diseases, biodiversity conservation, and health equity, which require coordinated efforts across countries and regions (Chen et al., 2019; Freshwater et al., 2006; Lucas, 2005).

International research collaborations can be categorized by the income level of partner countries. North–South collaborations involve partnerships between high-income countries (HICs) and low- and middle-income countries (LMICs), whereas South–South collaborations occur exclusively among LMICs. The latter have increasingly been recognized for their potential to strengthen local research capacity, promote contextually relevant knowledge production, and facilitate the exchange of experiences among countries facing similar socioeconomic and health challenges (Velasquez, 2024). Triangular collaborations involve simultaneous partnerships among HICs and other LMICs, with the aim of leveraging complementary resources, expertise, and perspectives.

Despite its benefits, these different collaboration models can be shaped by persistent asymmetries. Traditional North–South dynamics often mirror deeper inequalities in scientific infrastructure, funding, and epistemic authority. Previous analyses have shown that researchers from LMICs are less likely to occupy key authorship positions in international collaborations, even when the research is conducted in their own countries and especially when coauthors are from HICs (González-Alcaide et al., 2017; Hedt-Gauthier et al., 2019; Vecchio et al., 2025). Bibliometric evidence also shows that epistemic presence in the global science system (as measured by first authorship and citation impact) correlates with national investment in science, perpetuating structural advantages for high-income countries (Chinchilla-Rodríguez et al., 2019). Recent large-scale analyses further confirm that this hierarchical division of scientific labor is a persistent structural feature of contemporary international collaboration (Miao et al., 2024).

The sociology of science literature has long conceptualized these asymmetries through a Centre–Periphery model, in which scientific resources, prestige, and agenda-setting capacity remain concentrated in a small number of high-income countries, while peripheral nations risk incorporation into global networks on unequal terms (Schott, 1991). From this perspective, international collaboration may function either as a vehicle for genuine knowledge transfer and capacity-building, or as a mechanism that reinforces existing hierarchies, a tension that Academic Dependency Theory frames as the risk of peripheral countries trading scientific autonomy for visibility (Alatas, 2003; Leydesdorff & Wagner, 2008).

This model has nonetheless been subject to important critiques, as it risks portraying peripheral countries as structurally passive, underestimating the agency of emerging scientific powers and the self-organizing dynamics of global collaboration networks (Wagner & Leydesdorff, 2005). Bibliometric evidence shows that the relationship between international collaboration, leadership, and citation impact is not uniform across countries: nations with intermediate scientific capacity may derive citation advantages

from collaboration without assuming leadership roles, while simultaneously acting as regional knowledge hubs in partnerships with countries of lower scientific capacity (Chinchilla-Rodríguez et al., 2019). Whether such context-dependent positioning reflects genuine scientific autonomy or a more nuanced form of structural dependency remains an open empirical question that this study addresses in the Brazilian case.

Brazil, as Latin America's largest economy and scientific leader, responsible for 45.3% of the region's Scopus-indexed journals (Céspedes, 2021) and 3.2% of global public health output (Zacca-González et al., 2014), offers a compelling case study in developing international research strategies. Brazil's research system has grown substantially over the past three decades, driven by federal investment in graduate education and anchored in a network of federal and state universities and public research institutes, albeit with uneven distribution of research capacity across institutions and regions (Sidone et al., 2017). Historically embedded in a North–South collaboration model, Brazil has increasingly positioned itself as a regional leader in South–South and Triangular cooperation, engaging with both HIC and LMIC to bridge global scientific efforts, a trend noted in development cooperation discussion papers (Chaturvedi, 2011; Waisbich & Haug, 2022). This shift reflects a commitment to regional integration and development-oriented research aligned with the United Nations Sustainable Development Goals (SDGs), particularly in the HLS (McManus et al., 2023a, b). Policy-oriented analyses of South–South and triangular cooperation describe Brazil as a strategic intermediary between scientifically developed and developing nations. These accounts emphasize Brazil's role in facilitating more inclusive and horizontal forms of international collaboration that address shared challenges and local priorities (Ramírez, 2019; Waisbich & Haug, 2022).

Despite Brazil's growing visibility in global science, systematic analyses of the structure and evolution of its international collaborations in the HLS remain scarce. Prior studies have offered broad overviews of Brazilian science (Leta & Chaimovich, 2002; McManus et al., 2020), but often overlook the distinct collaboration models and variations in scientific leadership and thematic focus within them. Understanding these variations is critical for three reasons: (i) different collaboration models may yield different benefits in terms of capacity building and knowledge transfer; (ii) authorship patterns reflect power dynamics that influence research agendas and benefit distribution; and (iii) thematic specialization reveals whether collaborations address locally relevant priorities or reproduce external agendas.

This study fills that gap by conducting a bibliometric analysis of Brazil's international scientific collaborations in HLS between 2010 and 2019. We analyze North–South, South–South and Triangular partnerships, comparing their growth, authorship dynamics, citation impact, and thematic specialization. We aim to contribute to a better understanding of how an emerging scientific power engage in global collaboration networks, balancing leadership, impact and regional relevance. Specifically, we address four research questions: (i) How have the volume and composition of Brazil's international collaborations in HLS evolved across different collaboration models (North–South, South–South, Triangular)? (ii) To what extent does Brazil exercise scientific leadership within these models? (iii) How does citation impact vary across collaboration models? (iv) What thematic specializations characterize each model, and how do they align with Brazilian scientific expertise?

Method

Data selection, extraction, and processing

This study is based on bibliometric data retrieved from the Scopus database (Elsevier B.V.) using the Scopus Search API. The dataset includes original articles and review papers published between 2010 and 2019 with at least one author affiliated with a Brazilian institution. The pybliometrics library (Rose & Kitchin, 2019), was used to automate data extraction and caching. The initial dataset comprised 542,842 records from the HLS broad subject area, defined according to the All Science Journal Classification Codes (ASJC) available from Scopus and used to classify journals into a hierarchical structure across four broad subject areas: Life Sciences, Physical Sciences, Health Sciences, and Social Sciences and Humanities. The HLS broad subject area includes 11 subareas, 5 of them related to Life Sciences (Agricultural and Biological Sciences; Biochemistry, Genetics and Molecular Biology; Immunology and Microbiology; Neuroscience; Pharmacology, Toxicology and Pharmaceutics), 5 of them related to Health Sciences (Medicine; Nursing; Veterinary; Dentistry; Health Professions) and a Multidisciplinary area common in both.

The following query structure was applied: [SUBJAREA (medi OR nurs OR vet OR dent OR heal OR mult) OR SUBJAREA (agri OR bioc OR immu OR neur OR phar)] AND [AFFILCOUNTRY (brazil OR brasil) AND PUBYEAR > 2009 AND PUBYEAR < 2020] AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "re")).

The records were deduplicated by removing entries with identical Scopus EIDs and excluding documents with missing author affiliations. The final dataset consisted of 362,860 unique publications and was stored in an SQLite database to support structured querying and analysis. The VantagePoint software (Search Technology, Inc.) was used to standardize countries' names.

The same corpus was also queried in OpenAlex via DOIs to retrieve normalized citation indicators (see "Citation analysis" section) and to cross-validate Scopus-derived authorship records.

The analysis ended before 2020 to avoid the confounding effects introduced by the COVID-19 pandemic. The global health crisis altered research priorities and international collaboration dynamics, especially in the HLS, resulting in atypical publication patterns and a surge in pandemic-related outputs. Including post-2019 data would likely bias the analysis, obscuring long-term structural trends in collaboration and thematic focus. By focusing on the pre-pandemic decade, this study provides a more stable and representative picture of Brazil's evolving role in global scientific networks, avoiding crisis-driven distortions.

Specialization Index

The Specialization Index (SI; Aksnes et al., 2014) was used to measure the relative research participation by subject areas as compared to the global scientific productivity. It was calculated as the ratio of the Brazilian share of scientific output to the corresponding global share of the same area over the same period. A SI equal to 1 means Brazil's specialization matches the global average. Values > 1 indicate that the country is more specialized than

the global average, while values < 1 indicate less specialization than the global average in each subject area.

Classification of collaboration types

Only publications with at least one author affiliated to a country other than Brazil were included in the collaboration analysis; publications authored exclusively by Brazilian-affiliated researchers were excluded by definition. Co-authorship with authors affiliated to foreign countries was used as a proxy for international collaboration. Author affiliations were categorized by country income level according to the World Bank classification (World Bank, 2020), which divides economies into four income groups based on gross national income (GNI) per capita. Following established convention in the literature (Hedt-Gauthier et al., 2019; Vecchio et al., 2025), Low Income, Lower-Middle Income, and Upper-Middle Income economies were aggregated into a single Low- and Middle-Income Countries (LMIC) category and contrasted with High-Income Countries (HIC). This operationalization directly captures the income-based asymmetries central to North–South collaboration dynamics.

Collaborations were grouped into three categories: (i) BR–HIC: publications with at least one Brazilian affiliation and one or more affiliations exclusively from HIC (North–South); (ii) BR–LMIC: publications with Brazilian and exclusively LMIC affiliations (South–South); and (iii) BR–HIC–LMIC: publications involving at least one affiliation each from Brazil, HICs, and LMICs (Triangular).

Authors with multiple institutional affiliations spanning different country income categories were handled differently depending on the type of analysis. For collaboration type classification, publications were assigned to collaboration categories (BR–HIC, BR–LMIC, or BR–HIC–LMIC) based on the full set of unique country affiliations present among all authors, regardless of multiple affiliations. For example, if any author listed both a Brazilian and a Spanish institution, both countries were counted, resulting in the publication being classified as BR–HIC (or BR–HIC–LMIC if other LMIC affiliations were also present). For leadership analysis, authors with multiple affiliations including at least one Brazilian institution were classified as “Brazilian-affiliated” for the purpose of determining authorship position.

Leadership analysis

Authorship order was used as a proxy for scientific leadership, consistent with established practice in HLS where the first author is typically the main contributor and the last author is often the senior supervisor or principal investigator (Abramo et al., 2013a, 2013b). Specifically, we defined: (i) Exclusive Brazilian leadership: a Brazilian-affiliated author listed as both first and last author; (ii) Shared Leadership: a Brazilian-affiliated author listed as either first or last author (but not both); (iii) Non-Brazilian Leadership: neither the first nor last author were affiliated with a Brazilian institution.

To assess the potential influence of hyperauthorship on this indicator, we examined the distribution of author counts across the corpus: 97% of publications had 25 or fewer authors, with only 3% exceeding this threshold, indicating that hyperauthorship is unlikely to materially distort the leadership classification reported here.

Citation analysis

Field-normalized citation indicators were not available through standard Scopus exports and require access to proprietary modules such as SciVal. OpenAlex (Priem et al., 2022) was therefore used as a complementary, open-access source for these indicators. The same corpus of Brazilian publications retrieved from Scopus was matched against OpenAlex via DOI. Of the 98,680 international collaboration publications, 95,231 were successfully matched, yielding a retrieval rate of 96.5%, with no systematic decline in match rates across earlier publication years.

Two normalized citation indicators were evaluated. The “Outputs in Top Citation Percentiles” measures the share of publications among the most cited worldwide within their respective fields and years, as defined by OpenAlex. In this study, we adopted the top 10% threshold, a common benchmark in research evaluation. A higher proportion indicates greater research visibility and scholarly influence.

The “Field-Weighted Citation Impact” (FWCI) compares the number of citations received by a set of Brazilian publications with the average number of citations received by similar publications (same year, subject area, and document type). Herein an FWCI of 1.00 indicates that the Brazilian publications have been cited at the expected global average, while values above 1.00 indicate higher-than-average citation impact, and values below 1.00 indicate lower-than-average impact (SciVal-Elsevier, n.d.).

Thematic mapping

Term maps were created using VOSviewer 1.6.18 software (Centre for Science and Technology Studies, Leiden University, The Netherlands). Prior to term extraction, the textual content underwent a systematic and reproducible preprocessing pipeline designed for large-scale bibliometric co-occurrence mapping of Scopus corpora in the LHS. The pipeline was developed to address common limitations of broad biomedical literature exports, including encoding artefacts, multilingual leakage, out-of-domain contamination and critical polysemy in domain-specific terminology.

For each publication, title and abstract fields were processed through five sequential steps. First, encoding repair procedures were applied to correct common Latin-1 and CP1252 artefacts and recover misencoded characters. Second, structural cleaning was performed to remove HTML markup, URLs, DOI strings, in-text citation patterns, *p*-values, standalone years, percentages and other non-informative textual elements that could inflate irrelevant term co-occurrences. Third, a curated biomedical thesaurus was applied before tokenisation, using more than 60 regular-expression rules to normalise spelling variants, standardise acronym forms, disambiguate polysemous terms and preserve relevant multiword expressions as protected compound tokens. Fourth, tokenisation and lemmatisation were performed using the NLTK WordNetLemmatizer in two sequential passes, first as nouns and then as verbs, while thesaurus-generated compound tokens were preserved without lemmatisation. Finally, stopword and language filtering combined NLTK English stopwords, bibliometric stopwords, multilingual stopword lists and an English dictionary filter to reduce generic scholarly vocabulary, residual non-English terms and out-of-domain noise.

The resulting clean, space-separated token strings were stored as additional fields in the dataset and formatted as a VOSviewer-compatible corpus for downstream term extraction

and co-occurrence analysis. The full preprocessing workflow is publicly available in a GitHub repository, supporting reproducibility and reuse in similar bibliometric studies: <https://github.com/GRID-saude/Pipeline-For-Text-Data-Pre-processing>.

Terms were extracted from the pre-processed titles and abstracts and clustered according to the association strength measure proposed by van Eck et al. (2006). To balance comprehensiveness with computational tractability and noise reduction, only terms that appeared at least 10 times were considered, corresponding to approximately 60% of the corpus. The higher the number of publications in which two terms appeared together, the stronger their association, and the closer they were placed on the map. Graphically, each term was represented by a circle whose diameter and label size were proportional to term frequency. Network visualization and clustering parameters were held constant across all three collaboration model maps (BR–HIC, BR–LMIC and BR–HIC–LMIC) to ensure comparability. Layout parameters followed the VOS algorithm with Attraction = 2 and Repulsion = 1. Clustering was performed using a resolution of 1.5 and a minimum cluster size of 50 terms. The use of identical settings across maps ensured that visual differences between collaboration models reflected structural variation in co-word patterns rather than artefacts of parameter choice.

Semi-automated thematic classification

AI-assisted classification

Thematic clusters generated by VOSviewer were classified into Scopus ASJC subject areas using the large language model Claude Sonnet (version claude-sonnet-4-20250514, Anthropic). For each cluster, the model received a structured input containing all terms and their occurrence counts. Classification was performed via the Anthropic Messages API with temperature = 0 to ensure deterministic and reproducible outputs. The model was instructed to: (i) weight terms proportionally to their occurrence frequency; (ii) assign between 1 and 3 ASJC codes per cluster, ranked by relevance and restricted to a predefined list of specific subject area codes, excluding parent/general categories unless no specific child code was applicable; (iii) distribute confidence scores across assigned codes proportionally to their thematic weight, with all scores summing to 1.0; and (iv) return output exclusively in structured JSON format, containing the cluster identifier, a concise thematic label, the ranked ASJC codes with confidence scores, and a brief justification grounded in the highest-frequency terms. Each cluster was classified independently in a single API call. No conversation history was passed between calls. The complete system prompt, model version, and API parameters are available as Supplementary Material (Supplementary Material S1).

ASJC classification as post hoc triangulation

Since ASJC codes are assigned at the journal level in Scopus (not at the article level) they do not capture the specific thematic content of individual articles. Thematic clusters were therefore constructed inductively from co-occurrence patterns in titles and abstracts, independent of any a priori subject classification. The subsequent alignment of inductively derived clusters with ASJC categories served as a post hoc triangulation step: convergence between the two systems was interpreted as external validation of cluster coherence.

Reliability assessment

To evaluate the reliability of the AI-assisted classification, we conducted an inter-rater agreement experiment (McHugh, 2012). Two independent domain experts and the AI system each classified, independently and without access to one another's responses, the 12 co-citation clusters. Each rater assigned a single ASJC code per cluster from the predefined list. For the AI, agreement was assessed separately for the highest-ranked suggestion (Rank-1) and for the full set of three ranked suggestions (Rank-3), the latter capturing cases in which the expert-preferred category appeared within the model's ranked output without necessarily being the top choice. Cohen's κ was calculated for all three rater pairs (Rater 1 $\times$ Rater 2, Rater 1 $\times$ AI, Rater 2 $\times$ AI) at two levels of granularity: the 4-digit ASJC subcategory code and the 2-digit major disciplinary area (obtained by grouping codes by their first two digits). Bootstrap 95% confidence intervals were computed with 10,000 resamples (seed = 42). A minimum threshold of $\kappa \geq 0.80$ was adopted as the criterion for classification validation.

Inter-rater agreement for the 12 clusters was assessed across three rater pairs. Human inter-rater agreement at the 4-digit ASJC subcategory level was $\kappa_1 = 0.733$ (95% CI 0.463–1.000; Substantial), with three discrepancies, all occurring within the same disciplinary group (2-digit ASJC prefix), yielding perfect agreement at the major-area level ($\kappa_{1g} = 1.000$; Almost Perfect). Agreement between each human rater and the AI was $\kappa_2 = \kappa_3 = 0.821$ (95% CI 0.549–1.000; Strong) at the highest-ranked suggestion (Rank-1), and $\kappa_{2t} = 1.000$ when the full set of three ranked suggestions (Top-3) was considered, indicating that the AI included the expert-preferred category within its top suggestions for all 12 clusters. At the major-area level, all three rater combinations achieved $\kappa = 1.000$ (Almost Perfect). These results are consistent with the validity of the AI-assisted classification at the disciplinary-domain level and suggest a high degree of convergence between human expert judgement and AI-assigned categories across all granularity levels examined. Thematic clusters identified with the semi-automated ASJC classification are shown in Table 1.

Thematic specialization by collaboration model

Thematic specialization across collaboration models (BR–LMIC, BR–HIC, and BR–HIC–LMIC) was analyzed using VOSviewer's relevance score function. For each collaboration model, VOSviewer calculates a relevance score for every term by comparing its frequency within that specific model to its overall frequency across the entire corpus. Terms with higher relevance scores appear more frequently within a given collaboration model than would be expected based on their overall frequency across all collaboration models, indicating thematic areas that are particularly characteristic of that model. Based on these scores, the most representative terms of each collaboration model were qualitatively interpreted to identify distinct thematic specialization patterns across South–South, North–South, and Triangular collaborations.

Results

Share and specialization of Brazilian publications in HLS

Between 2010 and 2019, a total of 362,860 publications in HLS were published with at least one author affiliated to a Brazilian institution. The corpus comprised 338,925 original

Table 1 Thematic clusters identified with AI-assisted ASJC classification

Cluster	ASJC subject categories (Rank 1)	Associated terms (e.g.)
1	Ecology	Forest, diversity, tree, genus, taxon
2	Epidemiology	Confidence interval, trial, eye, placebo, stroke
3	Public Health	Questionnaire, physical activity, odds ratio, infant, randomized controlled trial
4	Biochemistry	Compound, peptide, enzyme, oil, inhibitor
5	Molecular Biology	Expression, microRNA, genome, chromosome, cell line
6	Cellular and Molecular	Mouse, receptor, rat, activation, inflammation
7	Animal Science and Zoology	Diet, cow, feed, supplementation, milk
8	Orthopedics and Sports Medicine	Exercise, session, blood pressure, knee, player
9	Immunology	Virus, parasite, macrophage, interleukin, immune response
10	Dentistry (miscellaneous)	Implant, bone, tooth, restoration, fracture
11	Agronomy and Crop Science	Soil, seed, crop, biomass, cultivar
12	Microbiology	Strain, <i>Escherichia coli</i> , serotype, <i>Pseudomonas</i> , <i>Mycobacterium</i>

Clusters are numbered as generated by VOSviewer co-citation analysis. Major field and subject categories reflect the AI-assigned classification AS/C All Science Journal Classification (Scopus)

research articles and 23,935 reviews, distributed across the 11 ASJC subject categories, with the 5 most frequent being: Medicine (43.4%), Agricultural and Biological Sciences (35.2%), Biochemistry, Genetics and Molecular Biology (19.6%), Immunology and Microbiology (7.5%), Pharmacology, Toxicology and Pharmaceutics (6.3%). This represents an average of 4% of the global publications ($n=13,974,467$) in the same broad subject area and period.

Scientific output in HLS reveals distinct patterns of specialization and global contribution (Table 2). In the Health Sciences, Dentistry stands out as the most specialized field, accounting for 13.7% of global publications ($SI = 3.7$). Veterinary Sciences also show significant specialization, with 10.6% of global publications ($SI = 2.9$). In the Life Sciences, Agricultural and Biological Sciences emerge as key strengths, contributing with 7.0% of global publications ($SI = 1.9$).

Brazilian International Collaboration in HLS

Overall, 264,180 (72.8%) publications were authored exclusively by Brazilian institutions (Brazil-only), while 98,680 (27.2%) involved at least one foreign co-author (Brazil-International Collaboration, or BR-IC, Fig. 1A).

During this period, international collaborations exhibited substantially higher growth than publications with only national co-authorship. Brazil-IC publications increased from 5,435 in 2010 to 14,748 in 2019, representing a 1.71-fold increase, compared to a 0.63-fold increase for overall Brazilian HLS output and 0.36-fold for Brazil-only publications (Fig. 1B).

International collaboration rates varied significantly across HLS subject areas (Fig. 1C). Veterinary Medicine, Nursing, and Dentistry show lower levels of international collaboration, with Veterinary Medicine having only 16% of publications involving international partners, and Dentistry and Nursing with 24% and 17%, respectively. These numbers contrast with other fields, such as Neuroscience (38%), Biochemistry (37%), and Immunology and Microbiology (34%), which display higher international collaboration.

Collaboration models: Volume, Growth Trends, Leadership and Citation Impact

Among international collaborations, BR-HIC represented the majority ($n=76,276$; 77.3%), followed by BR-HIC-LMIC ($n=14,097$, 14.3%) and BR-LMIC collaborations ($n=8,307$, 8.4%, Fig. 2A).

Growth patterns varied by collaboration model (Fig. 2B). BR-HIC-LMIC collaborations grew from 593 publications in 2010 to 2,418 in 2019 (3.08-fold increase), the highest among all categories. BR-LMIC collaborations increased from 462 to 1,352 (1.93-fold), while BR-HIC collaborations, despite remaining numerically dominant, showed more modest growth from 4,380 to 10,978 (1.51-fold, Fig. 2B).

Brazilian leadership, whether exclusive or shared, was more frequent in BR-HIC (67%; $n=47,927$) and BR-LMIC (75%; $n=5,473$) collaborations, but considerably lower in BR-HIC-LMIC collaborations (31%; $n=3,722$, Fig. 3). The exclusive leadership by Brazilian researchers was similar in BR-HIC (37%; $n=37,129$) and BR-LMIC (38%; $n=2,885$) collaborations. In BR-HIC-LMIC collaborations, shared leadership was more common (19%; $n=2,357$) when compared with exclusive leadership of Brazilian researchers (11%; $n=1,365$, Fig. 3). In the BR-HIC-LMIC scenario, HICs are the most frequent

Table 2 Number of publications, share (%) and specialization index by HLS fields (2010–2019)

Broad subject area	Major subject area	Brazilian publications (<i>N</i>)	World publications (<i>N</i>)	Brazilian share (%)	Specialization Index—SI [95% CI]
Health Sciences	Dentistry	19,103	139,620	13.7	3.7 [3.67–3.79]
	Veterinary	22,640	212,936	10.6	2.9 [2.86–2.94]
	Nursing	17,983	404,269	4.4	1.2 [1.19–1.23]
	Health Professions	11,662	278,992	4.2	1.1 [1.12–1.16]
	Medicine	157,418	5,780,018	2.7	0.7 [0.74–0.75]
	Agricultural and Biological Sciences	127,771	1,837,308	7.0	1.9 [1.89–1.90]
Life Sciences	Immunology and Microbiology	27,144	696,699	3.9	1.1 [1.05–1.07]
	Neuroscience	16,615	603,406	2.8	0.8 [0.74–0.76]
	Pharmacology, Toxicology and Pharmaceuticals	23,005	819,548	2.8	0.8 [0.76–0.77]
	Biochemistry, Genetics and Molecular Biology	71,255	2,708,730	2.6	0.7 [0.71–0.72]

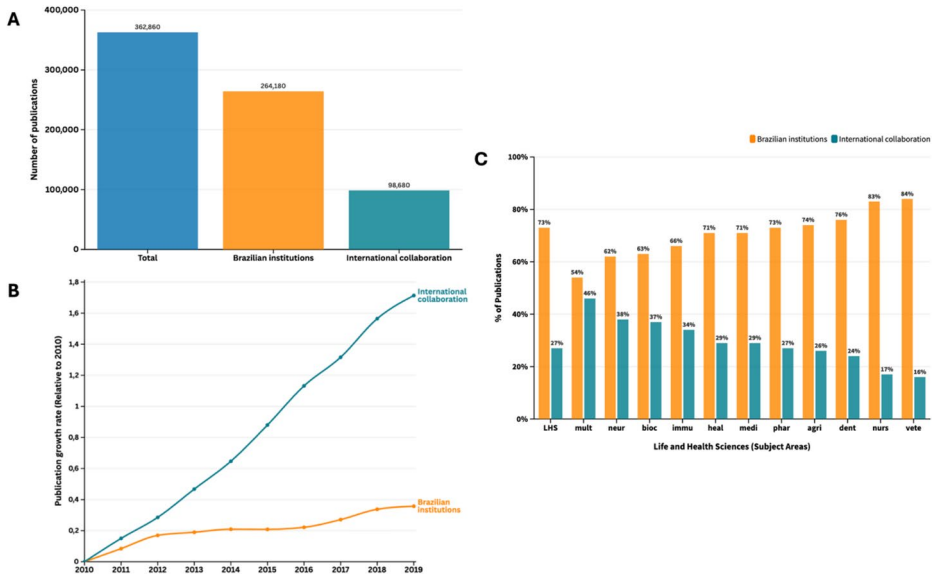


Fig. 1 Scientific publications in HLS involving Brazilian researchers (2010–2019). **A** Total number of publications. **B** Publication growth rate (%) relative to 2010. **C** International collaboration by HLS subject area. *Brazil-IC* Brazil-International Collaboration, *LHS* All publications, *AGRI* agricultural and biological sciences, *BIOC* biochemistry, genetics, and molecular biology, *IMMU* immunology and microbiology, *NEUR* neuroscience, *PHAR* pharmacology, toxicology, and pharmaceuticals, *MEDI* medicine, *NURS* nursing, *VETE* veterinary, *DENT* dentistry, *HEAL* health professions, *MULT* multidisciplinary sciences

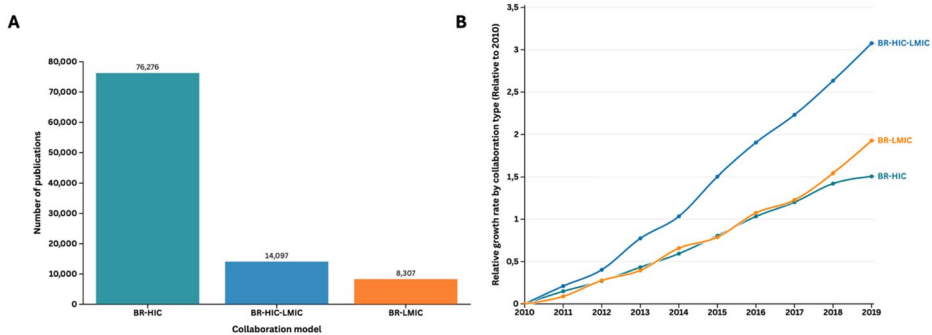


Fig. 2 Internationally collaborative scientific publications profile in HLS involving Brazilian researchers (2010–2019). **A** Total number of publications by collaboration model. **B** Relative growth rate of each collaboration type. *BR-HIC* at least one Brazilian affiliation and one or more affiliations exclusively from HIC, *BR-LMIC* Brazilian and exclusively LMIC affiliations, *BR-HIC-LMIC* at least one affiliation each from Brazil, HICs, and LMICs, *HIC* high-income country, *LMIC* low- and middle-income country

leaders (43%, $n=5,204$), followed by shared leadership with 30% ($n=3,622$) and LMICs leadership with 27% ($n=3,333$, data not shown).

Analysis of citation impact showed a clear hierarchy across collaboration models (Fig. 4A). Triangular collaborations (BR-HIC-LMIC) achieved the highest median FWCI (2.47; IQR 1.11–5.69), followed by BR-HIC (median = 1.64; IQR 0.73–3.34) and

BR Leadership Shared Leadership

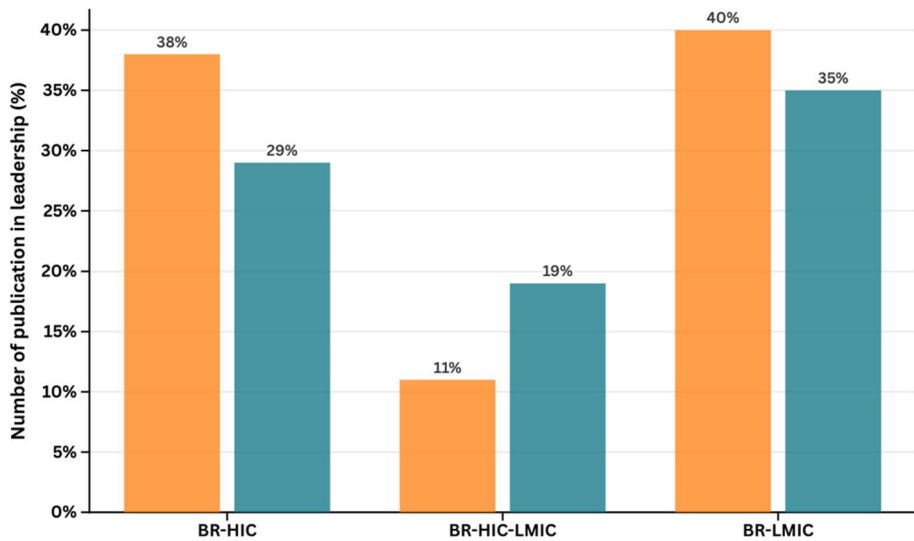


Fig. 3 Brazilian leadership for the three collaboration models. *BR-HIC* at least one Brazilian affiliation and one or more affiliations exclusively from HIC, *BR-LMIC* Brazilian and exclusively LMIC affiliations, *BR-HIC-LMIC* at least one affiliation each from Brazil, HICs, and LMICs, *BR leadership* Brazilian-affiliated researcher as first and last author, *Shared leadership* Brazilian-affiliated researcher as first or last author, *HIC* high-income country, *LMIC* low- and middle-income country

BR-LMIC (median = 1.02; IQR 0.39–2.17). These differences were statistically significant (Kruskal–Wallis $H = 2,916.13$, $p < 0.001$), and all pairwise comparisons remained significant after Bonferroni correction (Dunn post hoc test, all $p < 0.001$). The median-based analysis was preferred over means given the highly skewed distribution of FWCI values across all groups.

The proportion of publications ranked in the top 10% most-cited also differed significantly across collaboration models ($\chi^2 = 1,689.48$, $df = 2$, $p < 0.001$; Fig. 4B). BR-HIC-LMIC collaborations had the highest share, with 50.4% of publications ($n = 6,882$) in the top 10% most-cited, followed by BR-HIC (36.0%; $n = 26,564$) and BR-LMIC (23.2%; $n = 1,755$). Among top 10% publications, FWCI distributions remained significantly higher than non-top 10% publications within each group (Mann–Whitney U test, all $p < 0.001$): BR-HIC-LMIC showed the highest median FWCI within the top 10% stratum (median = 5.16), compared to BR-HIC (median = 3.76) and BR-LMIC (median = 3.07). In collaborations involving exclusively Brazilian affiliations, 46.1% of the publications (110,222) have an FWCI equal to or above 1 (data not shown).

Geographic patterns of collaboration

The United States (US) and the United Kingdom (UK) were the most frequent partners in both BR-HIC and BR-HIC-LMIC collaborations (Table 3). In the BR-HIC category, the US accounted for 43.7% of all publications, followed by the UK (12.5%), Spain (9.0%), and Canada (9.0%). In triangular collaborations (BR-HIC-LMIC), the US again led (56.4%), with the UK (27.7%) and Argentina (20.7%) among the top partners.

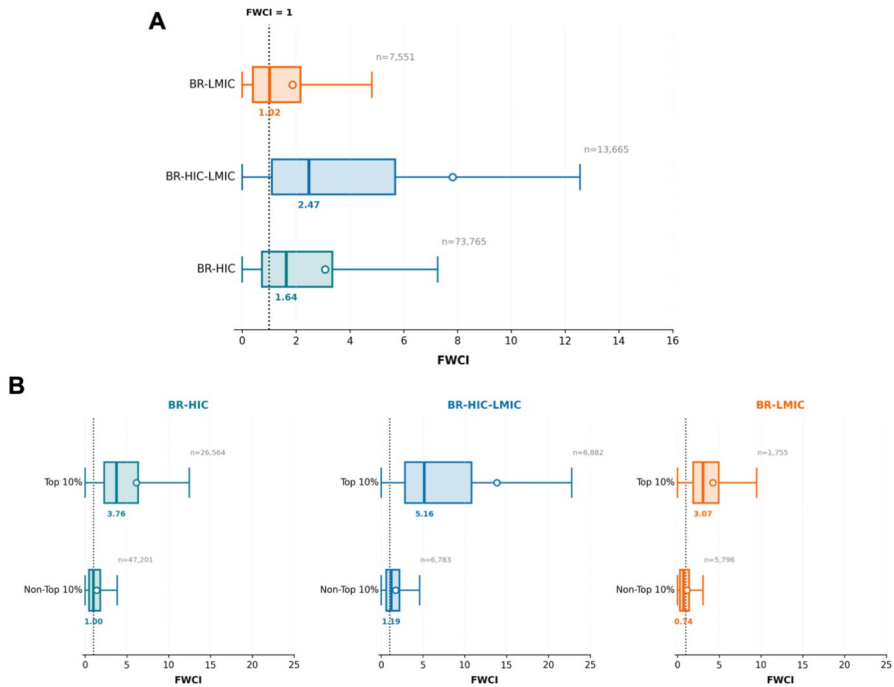


Fig. 4 Distribution of Field-Weighted Citation Impact (FWCI) and proportion of top 10% most-cited publications by collaboration model (2010–2019). **A** FWCI distribution across all publications for each collaboration model. **B** FWCI distribution stratified by citation impact stratum (Top 10% most-cited vs. Non-Top 10%) within each collaboration model. In both panels, boxes represent the interquartile range (Q1–3); vertical lines indicate the median; circles indicate the mean; whiskers extend to $1.5 \times \text{IQR}$; outliers omitted for clarity. The dotted vertical line indicates FWCI = 1 (global citation average). *BR-LMIC* publications with at least one Brazilian affiliation and one or more affiliations exclusively from high-income countries, *BR-HIC-LMIC* publications involving at least one affiliation each from Brazil, high-income countries, and low- and middle-income countries, *HIC* high-income country, *LMIC* low- and middle-income country

In South–South collaborations (BR–LMIC), Argentina (27.5%) and Colombia (16.2%) were Brazil’s leading partners, followed by Mexico (13.3%), Cuba (5.9%), and Peru (5.5%). Notably, Argentina was the only LMIC among the top three partners in both BR–HIC–LMIC and BR–LMIC modalities.

Thematic differentiation across collaboration types

A co-occurrence analysis of 34,562 terms (60% of total terms) from titles and abstracts identified 12 major thematic clusters in BR-IC publications (Fig. 5A). The AI-assisted classification assigned each cluster to a subject area according to the ASJC scheme, as follows: Ecology (red); Epidemiology (green); Public Health (blue); Biochemistry (yellow); Molecular Biology (purple); Cellular and Molecular Neuroscience (turquoise); Animal Science and Zoology (orange); Orthopedics and Sports Medicine (brown); Immunology

Table 3 Top collaborating partner countries by collaboration model, ranked by co-publication volume

Collaboration type	Rank	Collaborating countries	Number of publications (%) [*]
BR–HIC (North–South)	1	United States	33,302 (43.7)
	2	United Kingdom	9,524 (12.5)
	3	Spain	6,863 (9.0)
	4	Canada	6,850 (9.0)
	5	Germany	6,426 (8.4)
BR–HIC–LMIC (Triangular)	1	United States	7,951 (56.4)
	2	United Kingdom	3,906 (27.7)
	3	Argentina	2,914 (20.7)
	4	France	2,834 (20.1)
	5	Spain	2,829 (20.1)
BR–LMIC (South–South)	1	Argentina	2,287 (27.5)
	2	Colombia	1,346 (16.2)
	3	Mexico	1,101 (13.3)
	4	Cuba	493 (5.9)
	5	Peru	457 (5.5)

^{*}Percentages were calculated based on the total number of publications in each collaboration type

(pink); Dentistry—Miscellaneous (blush); Agronomy and Crop Science (olive green); and Microbiology (gray).

Thematic focus varied across collaboration types (Fig. 5B–D). BR–HIC collaborations encompassed a broad range of topics, with lower emphasis on Ecology; Epidemiology and Biochemistry (Fig. 5B). In contrast, BR–HIC–LMIC collaborations were highly concentrated in Epidemiology (Fig. 5C), while BR–LMIC collaborations focused primarily on Biochemistry; Ecology; Agronomy and Crop Science; and Microbiology (Fig. 5D).

Discussion

This study provides a structural and thematic overview of Brazil’s participation in international HLS collaborations between 2010 and 2019. Four research questions were examined. Regarding the volume and composition of Brazil’s international collaborations (RQ1), North–South partnerships remained the most prevalent collaboration model throughout the study period, while triangular collaborations exhibited the highest growth rate, increasing more rapidly than both North–South and South–South collaborations. Concerning scientific leadership (RQ2), Brazilian researchers held majority leadership positions in both North–South and South–South collaborations. In contrast, triangular collaborations were characterized by a greater presence of leadership from HIC institutions, with Brazilian researchers more frequently occupying shared or non-leading positions. This pattern is referred to here as context-dependent leadership. With respect to citation impact (RQ3), the three collaboration models displayed a clear gradient: triangular collaborations achieved the highest citation impact, followed by North–South collaborations, while South–South collaborations received the fewest citations on average. Thus, the collaboration model associated with lower levels of Brazilian leadership corresponded to the highest citation

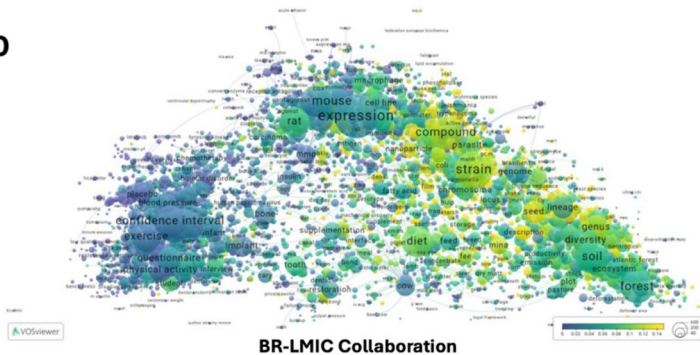
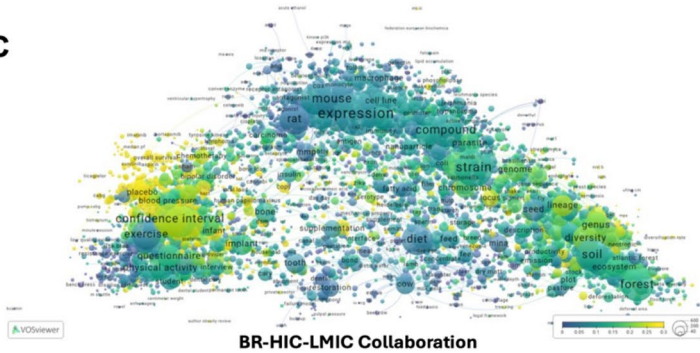
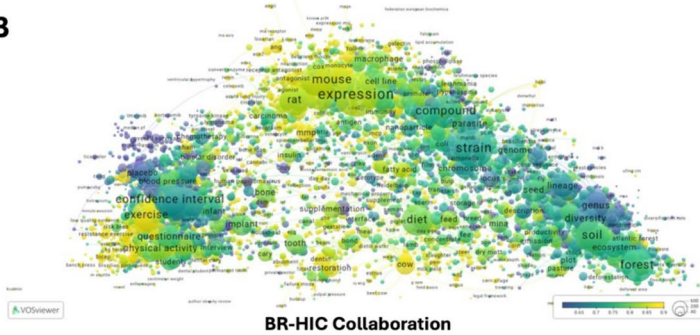
Fig. 5 Term co-occurrence maps in internationally collaborative scientific publications in HLS involving Brazilian researchers (2010–2019). **A** All BR-IC publications. Different colors distinguish between clusters of terms that have co-occurred more frequently. **B–D** Shared research interests are illustrated by an overlay of the most common terms in collaborations **B** BR–HIC, **C** BR–HIC–LMIC and **D** BR–LMIC. Terms were colored according to their score for each type of collaboration, from blue (lowest score) to yellow (highest score). (Colour figure online)

impact, whereas the models in which Brazilian researchers more frequently occupied leadership positions showed lower citation performance. Finally, regarding thematic specialization (RQ4), each collaboration model exhibited a distinct topical profile. North–South collaborations covered a broad range of research areas; triangular collaborations showed a strong concentration in clinical trials; and South–South collaborations were more strongly represented in tropical agriculture, ecology, and veterinary sciences. The discussion that follows examines these findings in greater detail and relates them to the literature on international scientific collaboration and Brazil’s role in the global research system.

Brazil has established itself as a key contributor to the global HLS research landscape, with a consistent increase in output over recent decades (Tramontin et al., 2022). Between 1995 and 2019, the country’s presence in high-impact journals grew notably: from 0.3 to 1.5% in general health sciences and 1% to 3% in public health (Celeste et al., 2021). Brazil’s scientific specialization in Dentistry, Veterinary Sciences, and Agricultural and Biological Sciences reflects both strategic priorities and structural characteristics of the country. Brazil has the largest number of dentists in the world and had a rapid expansion of dental schools (Chaves et al., 2018). The implementation of the National Oral Health Policy in 2004, known as *Brasil Sorridente* (Smiling Brazil), likely influenced the national research agenda by stimulating studies on major oral health issues and promoting the development of new products and technologies to support the expansion of public dental services (Chaves et al., 2017). In Agricultural and Biological Sciences, Brazil’s specialization likely reflects the central role of agribusiness (which accounts for over 20% of the national gross domestic product and 40% of exports) combined with the country’s extraordinary biodiversity, which positions it as a globally relevant site for ecological, taxonomic, and biological research. Scientific research has been pivotal in enabling agriculture in tropical environments through the development of cultivars adapted to local climate and soil conditions, supported by sustained government investment since the creation of EMBRAPA (Brazilian Agricultural Research Corporation) in 1973. The prominence of Veterinary Sciences is likewise tied to agribusiness, as expanding meat production continues to drive demand for veterinary research.

Our findings show that Brazil’s international collaboration in HLS grew substantially between 2010 and 2019. Prior studies have already documented Brazil’s growing participation in global scientific output (McManus et al., 2020; McManus et al., 2023a; Santin et al., 2016). This trend reflects a broader shift toward internationalization in Brazilian science, consistent with global patterns and policy-driven efforts to enhance the country’s visibility and integration within international research networks (McManus et al., 2020). Despite these advances, Brazil’s level of international collaboration in the HLS (27.2%) remains below the average observed among countries with comparable scientific development (46.1% across all fields) (Chinchilla-Rodríguez, 2018).

Examining leadership patterns across collaboration models reveals that Brazil’s position in global research networks is not fixed but context-dependent: a majority leader in South–South, a leading yet asymmetric partner in North–South ones, and structurally subordinate in multi-income arrangements, where HIC institutions predominate. This



gradient of leadership constitutes a more nuanced picture than the simple peripherality often attributed to emerging scientific powers in the North–South literature and resonates with the theoretical tension at the heart of the Centre–Periphery model and its critiques. Rather than occupying a fixed position in the global science hierarchy, Brazil exhibits the context-dependent positioning that bibliometric evidence has increasingly documented for emerging scientific powers as neither fully peripheral nor consolidated as a centre (Chinchilla-Rodríguez et al., 2019).

The thematic patterns observed across collaboration models offer a complementary interpretation of this differentiated positioning. In North–South collaborations (BR–HIC), the broad thematic diversity is consistent with partnerships driven at least in part by access to research infrastructure, advanced methodological expertise, and competitive funding that remain unevenly distributed between high- and lower-income research systems. From this perspective, the somewhat lower Brazilian leadership in North–South (BR–HIC) collaborations relative to South–South (BR–LMIC) ones may reflect the complementary logic of these partnerships.

South–South collaborations (BR–LMIC), by contrast, show an alignment between thematic focus and Brazil’s recognized areas of scientific specialization. The predominance of Ecology and Zootechny & Nutrition in these partnerships corresponds to fields in which Brazil holds high SI Index values. This convergence is consistent with an interpretation in which Brazil’s leadership in South–South collaborations reflects the contribution of established domestic expertise to address shared challenges in tropical and subtropical contexts; problems that are systematically underrepresented in Northern-led research agendas. Taken together, these patterns suggest that Brazil leads where it has the most to offer and collaborates where it has the most to gain.

The growing trend in Triangular collaborations (BR–HIC–LMIC) was not accompanied by a proportional increase in Brazilian leadership, suggesting that Brazil’s integration into multi-income networks has expanded in volume without a corresponding gain in agenda-setting power. While our data capture structural co-authorship patterns rather than the intentionality behind collaborative arrangements, the combination of indicators observed (authorship position, thematic focus, and citation impact) allows us to propose structural hypotheses about Brazil’s functional role within these partnerships. The limited Brazilian leadership in BR–Triangular collaborations, combined with their predominant focus on Clinical Trials, raises the possibility that Brazil functions primarily as a research site, data provider, or regional recruitment hub rather than an intellectual or methodological leader. Although our bibliometric data do not allow direct assessment of benefit-sharing arrangements, the observed pattern is compatible with concerns raised in the literature on “scientific extractivism”, whereby LMIC institutions contribute patients, data, and infrastructure while occupying limited leadership positions. Recent bibliometric evidence suggests that such patterns are not isolated cases but reflect a broader and persistent hierarchy in international scientific collaboration. Researchers from countries with lower scientific capacity are systematically less likely to occupy leadership positions and are more likely to participate in internationally collaborative research that diverges from their domestic research priorities, limiting the extent to which they can shape scientific agendas and fully benefit from collaborative activities (Miao et al., 2024). This pattern may reflect HICs’ strategies to access diverse patient populations for multicentric clinical trials, or to use Brazil’s relatively established research infrastructure as a gateway to other Latin American contexts. This interpretation is consistent with evidence from global health research showing that LMIC researchers remain systematically underrepresented in first and last authorship positions, even in studies conducted in their own countries (Dimitris et al., 2021; Naidoo

et al., 2021). Clinical trials typically have clearly defined leadership hierarchies that may disadvantage LMIC (Alemayehu et al., 2018), raising concerns about unequal benefit-sharing, particularly in scientifically vulnerable contexts (Alemayehu et al., 2018; Grover et al., 2017). Additionally, Triangular partnerships may involve more complex coordination demands that favor HIC researchers, who often have greater institutional resources, and experience in managing international collaborations. Lower leadership rates in such contexts may thus reflect capacity constraints rather than deliberate exclusion.

Argentina's prominent role across collaboration models warrants specific attention. As the only LMIC appearing among the top three partners in both South–South (BR–LMIC) and Triangular (BR–HIC–LMIC) collaborations, Argentina occupies a structurally distinct position within Brazil's international network functioning simultaneously as a peer partner in South–South arrangements and as a regional bridge in Triangular collaborations. This dual presence suggests that within Latin American science, a small group of countries with relatively stronger research infrastructure may serve as nodes connecting South–South networks to broader international partnerships, a dynamic that merits further investigation at the regional level.

The superior citation performance of Triangular collaborations (BR–HIC–LMIC) may stem from at least three non-exclusive explanations, presented here as plausible interpretations rather than empirically tested mechanisms. First, the thematic concentration of Triangular collaborations in clinical trials may inflate citation metrics, as trials typically attract higher citation rates and are more often published in high-impact journals (Kulkarni et al., 2007). Second, these collaborations may benefit from a “visibility multiplier” effect: publications co-authored by researchers from multiple countries and income levels tend to be perceived as more internationally relevant, increasing their readership and citation potential. This aligns with prior evidence showing that papers involving a larger number of countries receive more citations on average (Royal Society, 2011). Third, the participation of HIC institutions in triangular partnerships may facilitate access to high-impact publication venues that are less accessible to bilateral South–South collaborations.

South–South (BR–LMIC) collaborations showed the lowest citation impact, likely reflecting topic specificity rather than research quality. Notably, Brazil's South–South partnerships concentrated in Zootechny & Nutrition; Taxonomy, Systematics & Ecology; and Applied Biochemistry are aligned with Brazil's recognized areas of specialization in Agricultural and Biological Sciences; and Veterinary Sciences. This pattern suggests that Brazil leverages its specialized knowledge to address shared challenges among tropical and sub-tropical LMICs. Zootechny and nutrition research (veterinary/agricultural expertise) optimizes livestock production under similar climatic conditions; taxonomy and ecology work (biological sciences expertise) catalogs biodiversity in tropical ecosystems often neglected by Northern research agendas; and applied biochemistry explores practical applications of regional biological resources. While locally critical and building on Brazilian strengths, these highly specialized topics attract limited global interest and citations, being relevant primarily to researchers facing similar regional contexts. Lower bibliometric impact thus reflects strategic South–South collaboration in Brazil's areas of excellence, addressing specific shared problems among LMICs rather than pursuing the broader questions that dominate international research agendas.

Although the 2010–2019 period was deliberately selected to avoid the confounding effects of the COVID-19 pandemic, the structural trends identified here can be partially contextualized against pandemic-era evidence. Research on international collaboration during 2020–2021 suggests that crisis conditions may have temporarily altered some of the asymmetries documented in pre-pandemic HIC–LMIC partnerships: shared leadership was

the predominant authorship pattern in HIC–LMIC COVID-19 publications, with LMIC researchers leading exclusively in a slightly higher proportion than HIC researchers (Carvalho et al., 2023). At the same time, the pandemic introduced broader disruptions to international collaboration, including a measurable reduction in international co-authorship propensity and a temporary contraction of collaborative networks (Abramo et al., 2022; Cai et al., 2021). Whether the structural patterns identified in this study (Brazil’s context-dependent leadership gradient, its thematic specialization in South–South partnerships, and its subordinate positioning in triangular arrangements) have been reinforced, disrupted, or reconfigured in the post-pandemic period remains an open empirical question that warrants dedicated investigation.

These findings suggest, though do not directly test, several implications for Brazilian science policy. The alignment between Brazil’s areas of scientific specialization and the thematic focus of its South–South collaborations suggests that funding agencies could strategically leverage existing domestic strengths to deepen bilateral partnerships with LMICs facing shared tropical and subtropical challenges. At the same time, Brazil’s limited leadership in Triangular collaborations points to a structural gap between participation and agenda-setting power. Policies aimed at strengthening Brazilian leadership in multi-income arrangements could help translate Brazil’s growing collaborative output into more equitable research partnerships. Finally, evaluation frameworks that rely solely on citation-based metrics risk systematically undervaluing South–South collaborations, whose scientific relevance is regional rather than global. Broadening assessment criteria to include thematic relevance and knowledge transfer would better capture the full contribution of Brazil’s collaborative strategy.

Limitations

While this study offers valuable insights into Brazil’s international scientific collaboration, a few considerations are worth noting. Scopus, a widely recognized and comprehensive database, provides a robust foundation for analyzing research output (Baas et al., 2020); however, it may not fully capture publications in regional, national, or non-indexed journals. Consequently, our findings may overestimate the internationalization of Brazilian HLS research by systematically excluding domestically-oriented publications that appear only in local outlets. The precise proportions of BR–HIC, BR–LMIC, and BR–HIC–LMIC publications may also be partly influenced by Scopus’s indexing criteria. Nevertheless, the predominance of BR–HIC collaborations seen herein is consistent with Brazil’s long-standing policy of fostering graduate training in the United States and Europe, which has promoted enduring scientific ties with HIC institutions since the 1950s (Rezende, 2010).

Our analysis is based on formal institutional affiliations, which, while methodologically consistent, may not reflect informal or unpublished collaborations that also contribute meaningfully to scientific exchange. Our leadership classification counted authors with both Brazilian and foreign affiliations as Brazilian-affiliated, which may overestimate Brazilian leadership in cases where researchers are primarily based abroad but maintain secondary Brazilian affiliations. Furthermore, our use of first and last authorship positions as a proxy for scientific leadership, while well-established and widely applied in the health and life sciences (Abramo et al., 2013a, 2013b), may not capture the full complexity of scientific contributions.

As with most bibliometric approaches, this study focuses on quantitative indicators and may not fully account for the qualitative impact or societal relevance of the research

examined. Also, citation impact comparisons across collaboration models may partly reflect the citation advantage associated with a larger number of collaborating countries, an effect not examined in this study.

Our “operational” bibliometric definition of Triangular collaborations does not assume intentional policy-driven cooperation, but rather captures a structural pattern of multi-income collaboration that may or may not reflect formal Triangular cooperation arrangements.

Finally, our analysis covers 2010–2019, deliberately excluding the COVID-19 pandemic period to avoid atypical patterns. While this decision enhances internal validity, it means our findings may not reflect current collaboration dynamics, particularly given that the pandemic substantially altered international research partnerships, funding priorities, and publication patterns in health sciences. Future research will examine whether trends identified here persisted, accelerated, or reversed during and after the pandemic.

Conclusions

By jointly analyzing volume, leadership, citation impact, and thematic specialization across three collaboration models, this study advances understanding of how emerging scientific powers navigate global research networks, moving beyond binary characterizations of peripherality and centrality. Rather than occupying a fixed position, Brazil’s role varies systematically across collaboration types, a finding that calls for more granular frameworks in the bibliometric analysis of international collaboration, ones that account for collaboration-type-specific dynamics rather than treating a country’s global position as uniform. The gradient of leadership observed here, shaped by distinct thematic logics in each collaboration model, suggests that authorship patterns, citation impact, and thematic specialization must be interpreted jointly to reveal the functional roles that countries play in global science.

For the study of South–South and Triangular cooperation specifically, our findings offer an empirical counterpoint to policy narratives that treat these models as straightforwardly horizontal or equitable. The structural patterns observed suggest that Triangular arrangements can reproduce hierarchical dynamics even when framed as instruments of equity. Bibliometric analysis of authorship and thematic focus offers a tractable method for monitoring whether the equity aspirations of triangular cooperation are realized in practice, complementing qualitative and policy-oriented approaches.

The patterns documented here invite a reconsideration of how country-level positioning in global science is conceptualized and measured. If a country’s role varies systematically across collaboration types, then aggregate indicators of internationalization or leadership risk obscuring more than they reveal. Future research should examine whether the context-dependent positioning observed for Brazil characterizes other emerging scientific powers, and whether the alignment between domestic specialization and South–South thematic focus represents a broader mechanism through which countries in the Global South exercise scientific agency. Developing collaboration-type-sensitive frameworks may be essential to capture the full complexity of how knowledge is produced, exchanged, and governed in an increasingly multipolar global research landscape.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11192-026-05781-0>.

Author contributions JFL and PCA wrote the initial manuscript draft and conducted the data analysis and visualization. BPF revised and edited the manuscript, coordinated the study and acquired funding. EA performed data collection and data curation. FZ contributed to the interpretation of findings. All authors reviewed and approved the final manuscript.

Funding The Article Processing Charge (APC) for the publication of this research was funded by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - Brasil (CAPES) (ROR identifier: 00x0ma614). This study was supported by Fundação Carlos Chagas Filho de Amparo à Pesquisa do Estado do Rio de Janeiro, SEI-260003/001550/2022.

Data availability The processed dataset supporting the findings of this study is available from the corresponding author upon request. Raw data were retrieved from Scopus (Elsevier B.V.) and are subject to licensing restrictions.

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Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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